ENTERPRISE SERVER DEPLOYMENT AND OPERATING SYSTEM INSTALLATION

Installation Guide

ITEP 414 — System Administration and Maintenance

Week 3 Portfolio Project

Bachelor of Science in Information Technology (BSIT)

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1. Project Overview

Operating systems form the foundation of every enterprise IT infrastructure. Before a server can host websites, databases, cloud services, or enterprise applications, it must first be properly installed, configured, secured, and documented.

This project places the author in the role of a Junior System Administrator at ABC Startup Solutions, tasked with deploying the company's first Ubuntu Server. The scope of work includes a complete operating system installation, configuration of essential settings, verification of server functionality, comparison of modern boot technologies (BIOS vs. UEFI), and creation of professional documentation suitable for future administrators.

This Installation Guide documents every stage of the deployment process and serves as a portfolio artifact demonstrating foundational System Administration competencies.

2. Objectives

Knowledge Objectives

- Explain the purpose of an operating system in enterprise environments.
- Differentiate BIOS and UEFI firmware.
- Explain the stages of the computer boot process.
- Compare Ubuntu Server, Windows Server, and Rocky Linux.

Skills Objectives

- Install Ubuntu Server in a virtual machine.
- Configure server settings during installation.
- Enable secure remote administration using SSH.
- Verify server functionality.
- Document installation procedures.
- Produce professional technical documentation.

3. Software Requirements

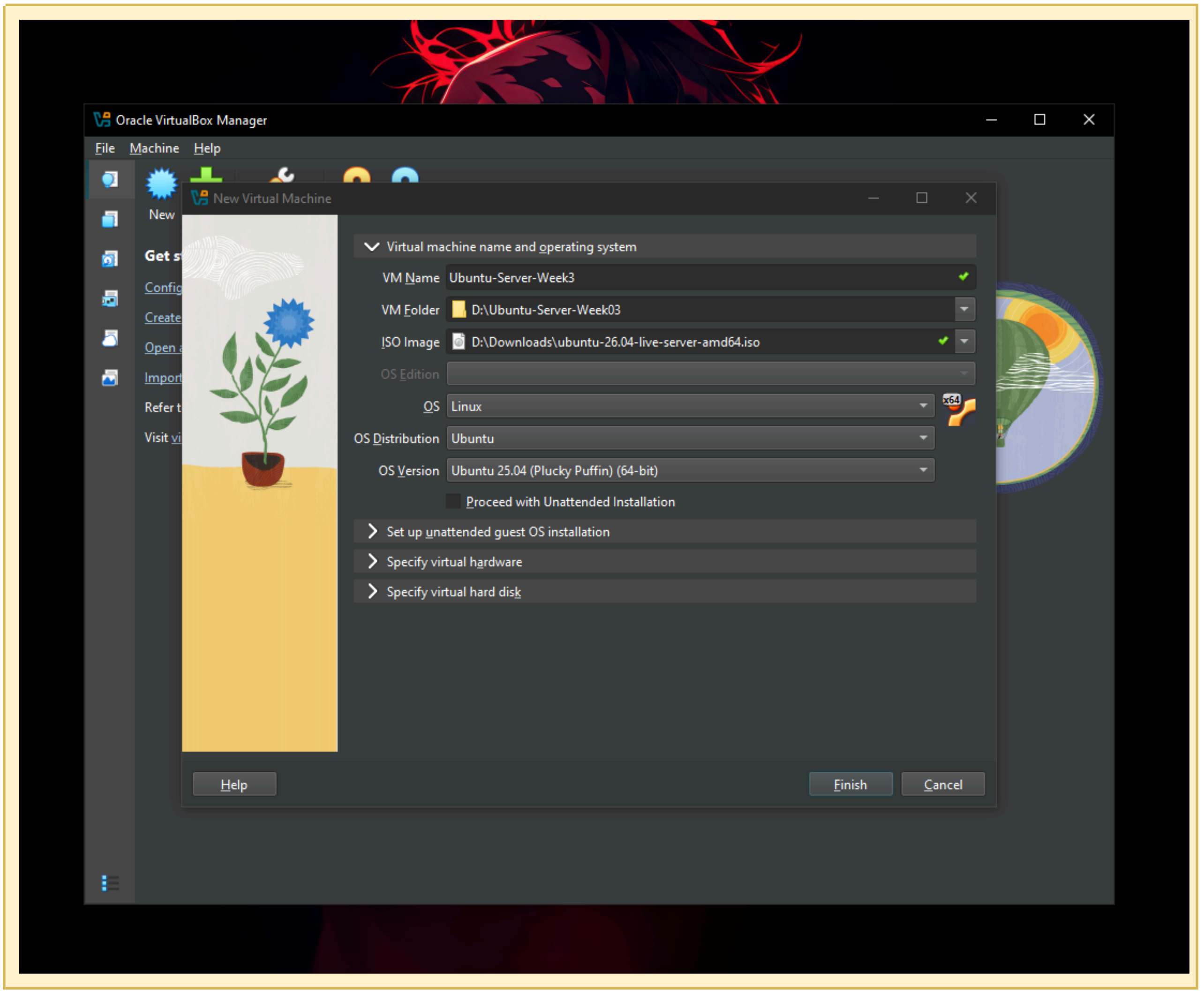
The following software was used to complete this deployment:

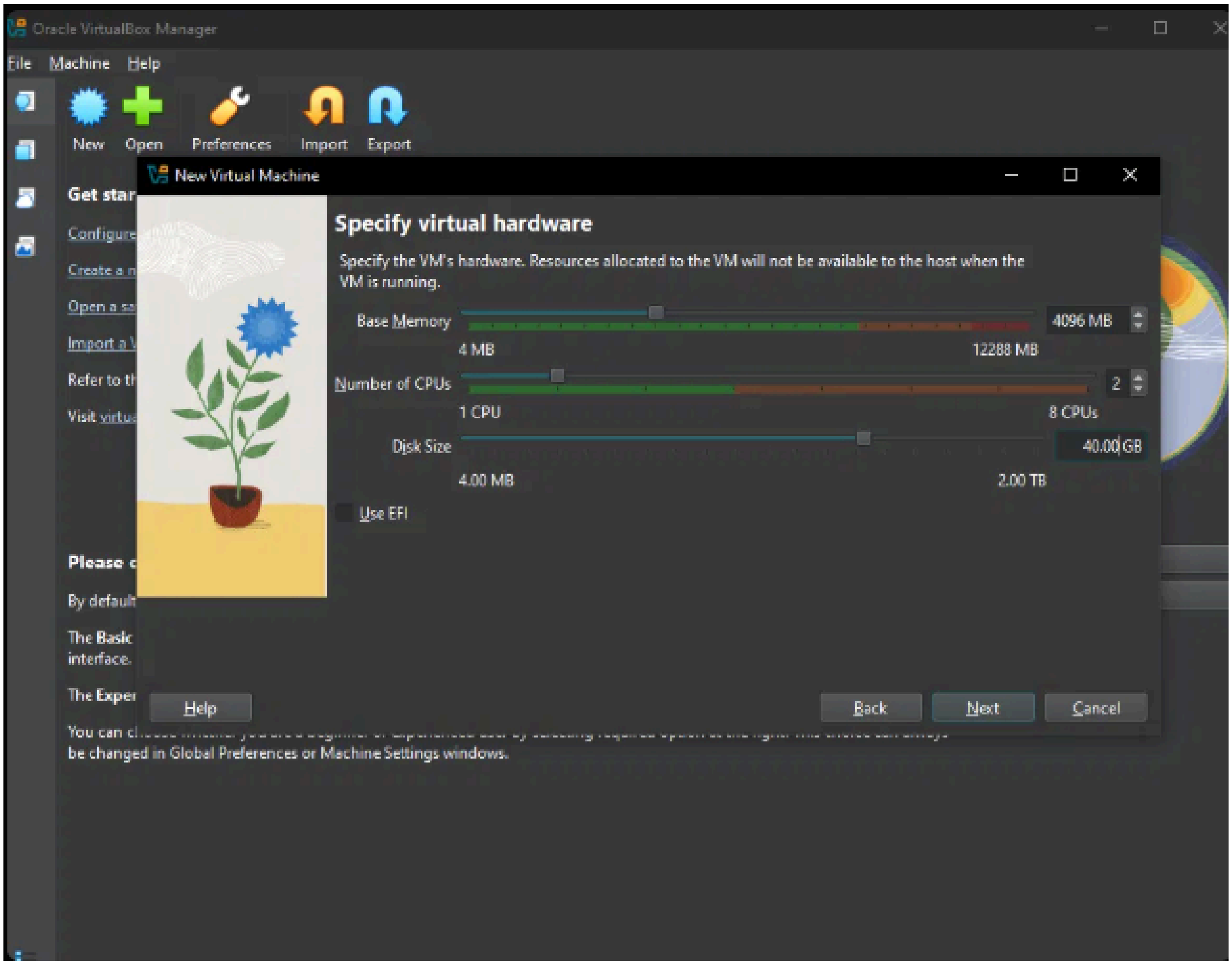
- Oracle VirtualBox / VMware Workstation — virtualization platform
- Ubuntu Server LTS ISO (latest stable version) — primary server operating system
- Windows Server Evaluation ISO — used for the bring-home comparison activity

4. Virtual Machine Specifications

The virtual machine was configured with the following minimum specifications as required by the project brief:

Component	Configured Value
Name	Ubuntu-Server-Week03
RAM	4 GB
CPU	2 Virtual Processors
Storage	40 GB (VDI/VMDK)
Network	NAT (or Bridged if instructed)
Optical Drive	Ubuntu Server ISO





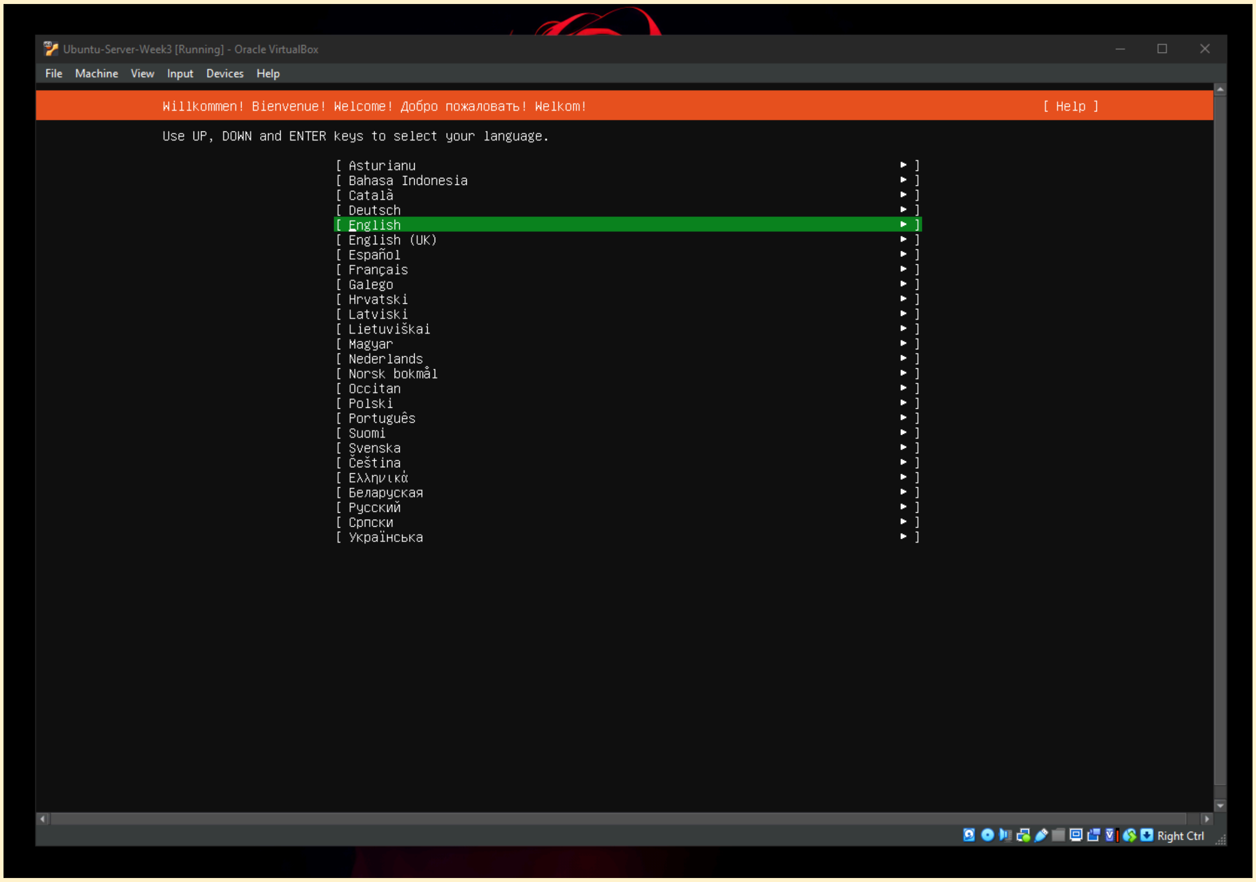
Oracle VirtualBox Manager
File Machine Help
New Open Preferences Import Export
New Virtual Machine
Specify virtual hardware
Specify the VM's hardware. Resources allocated to the VM will not be available to the host when the VM is running.
Base Memory
4096 MB
4 MB
12288 MB
Number of CPUs
2
1 CPU
8 CPUs
Disk Size
40.00 GB
4.00 MB
2.00 TB
Use EFI
Help
Back
Next
Cancel
be changed in Global Preferences or Machine Settings windows.

5. Step-by-Step Installation Procedure

The installation wizard was completed using the following configuration choices:

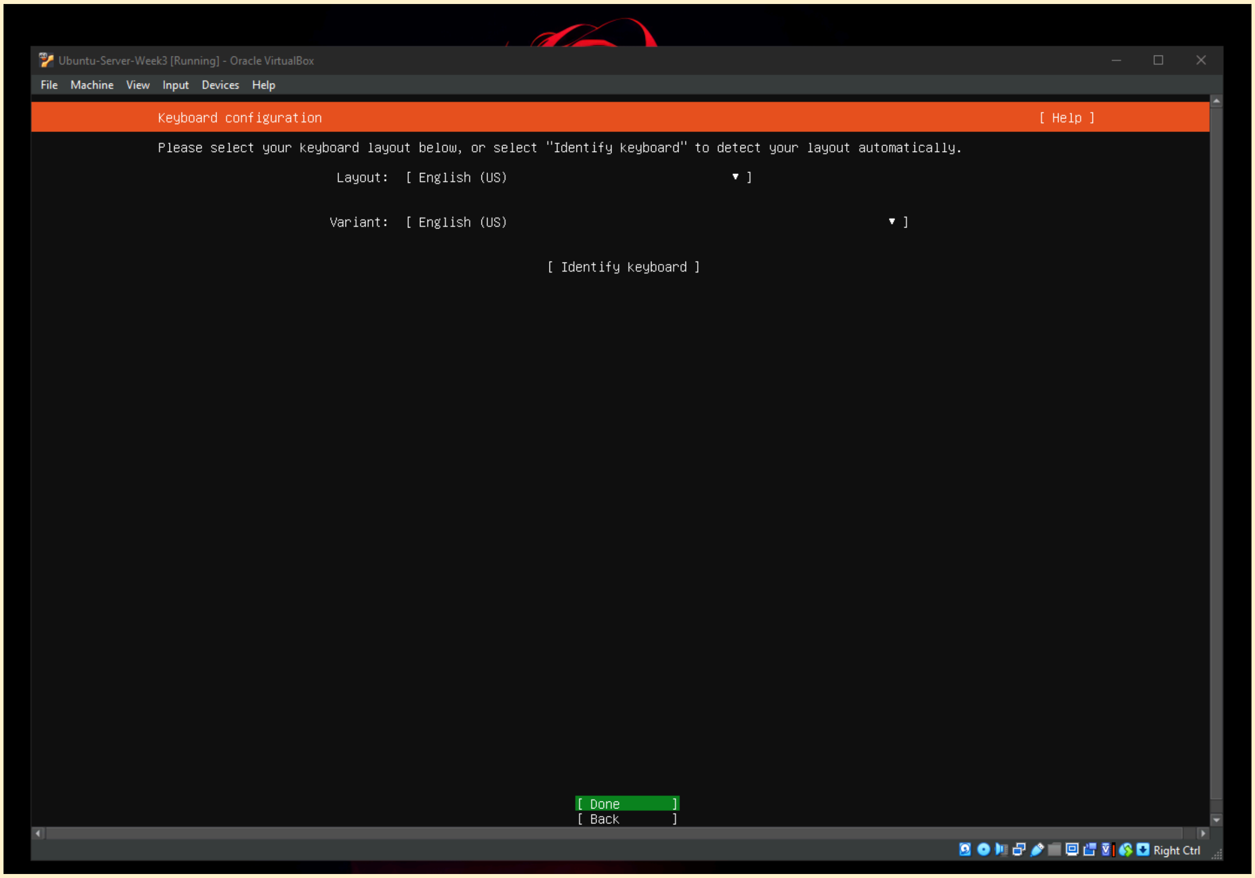
5.1 Language

Selected: English



5.2 Keyboard Layout

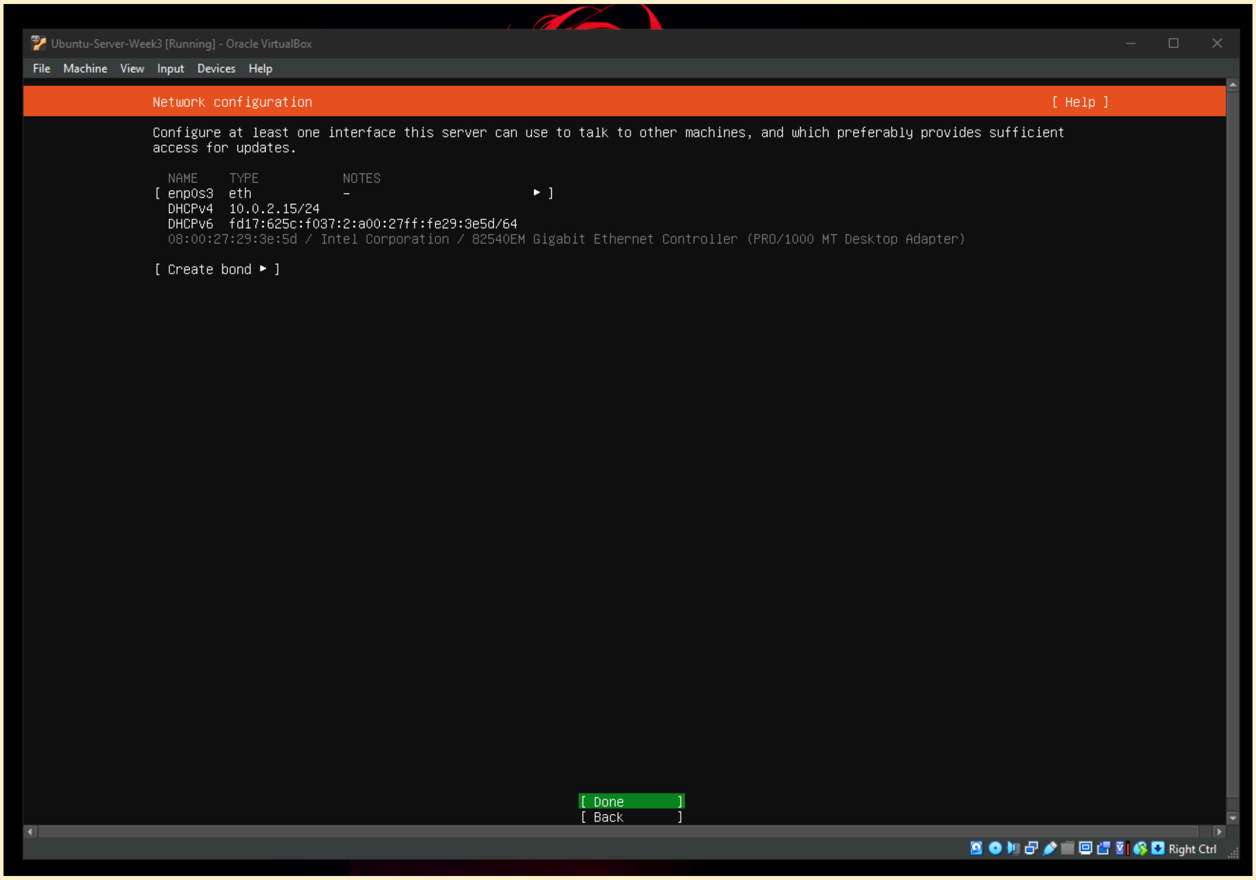
Selected: English (US)



5.3 Network Configuration

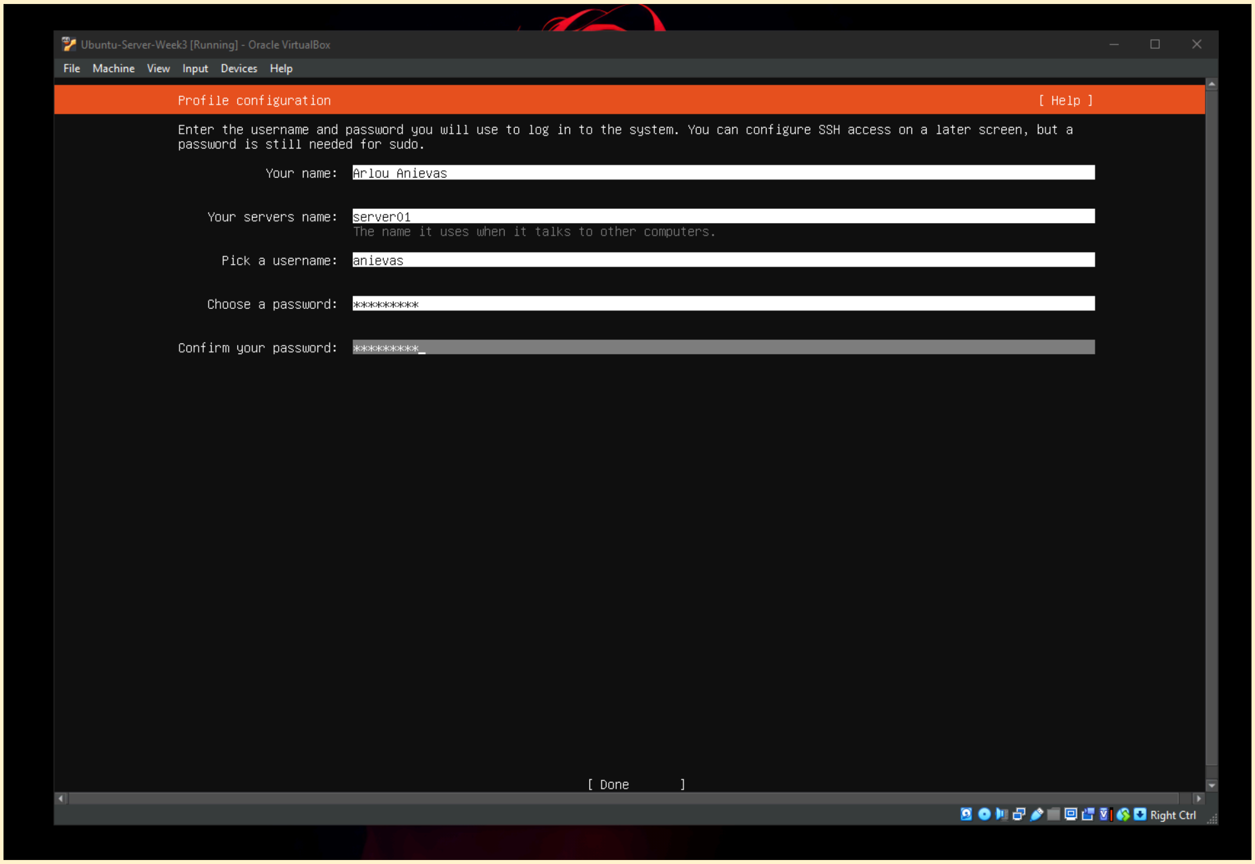
DHCP was accepted for automatic network configuration.

- Assigned IP Address: [enter IP address, e.g., 192.168.1.XX]



5.4 Hostname

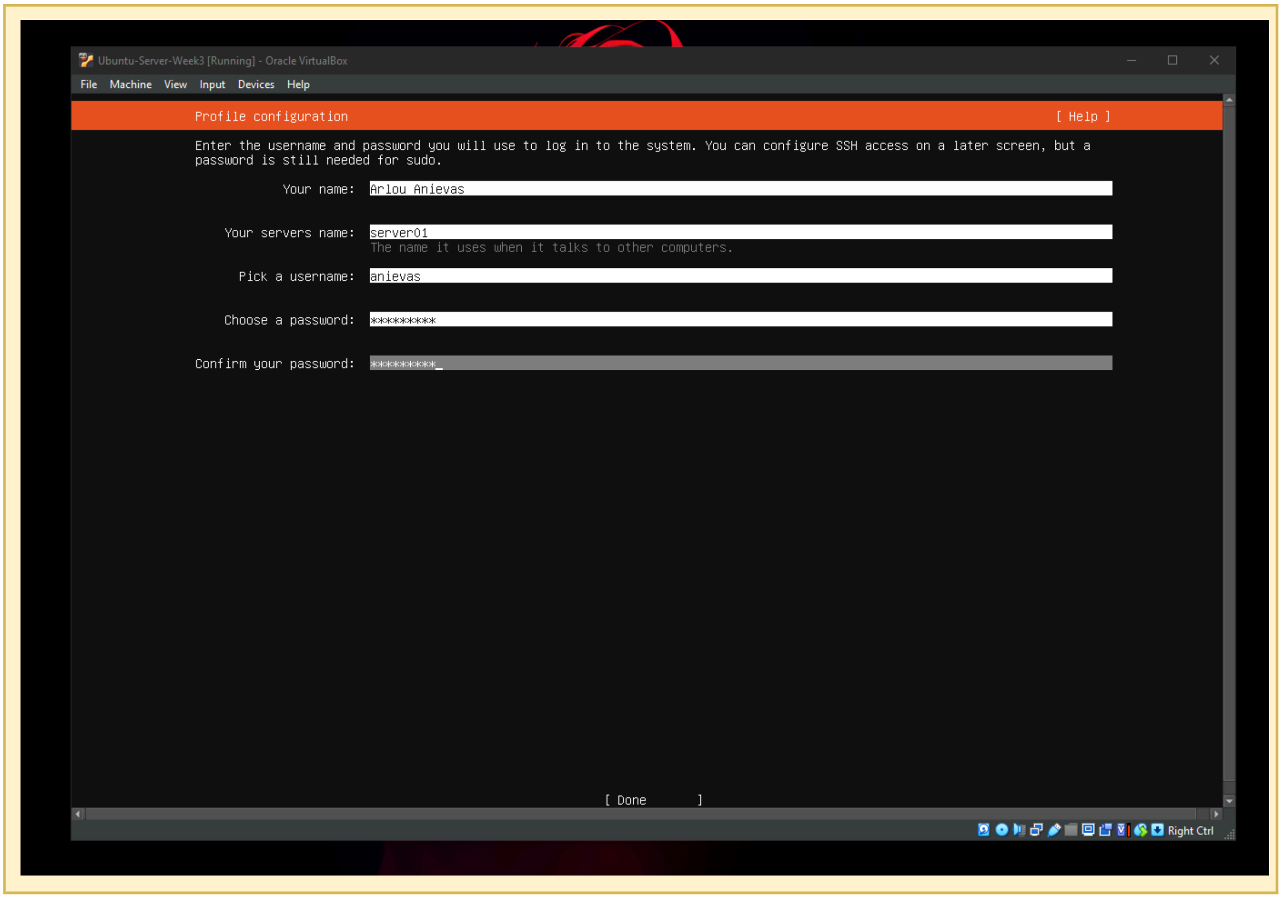
Assigned hostname: server01



5.5 User Account

A non-root administrative user was created:

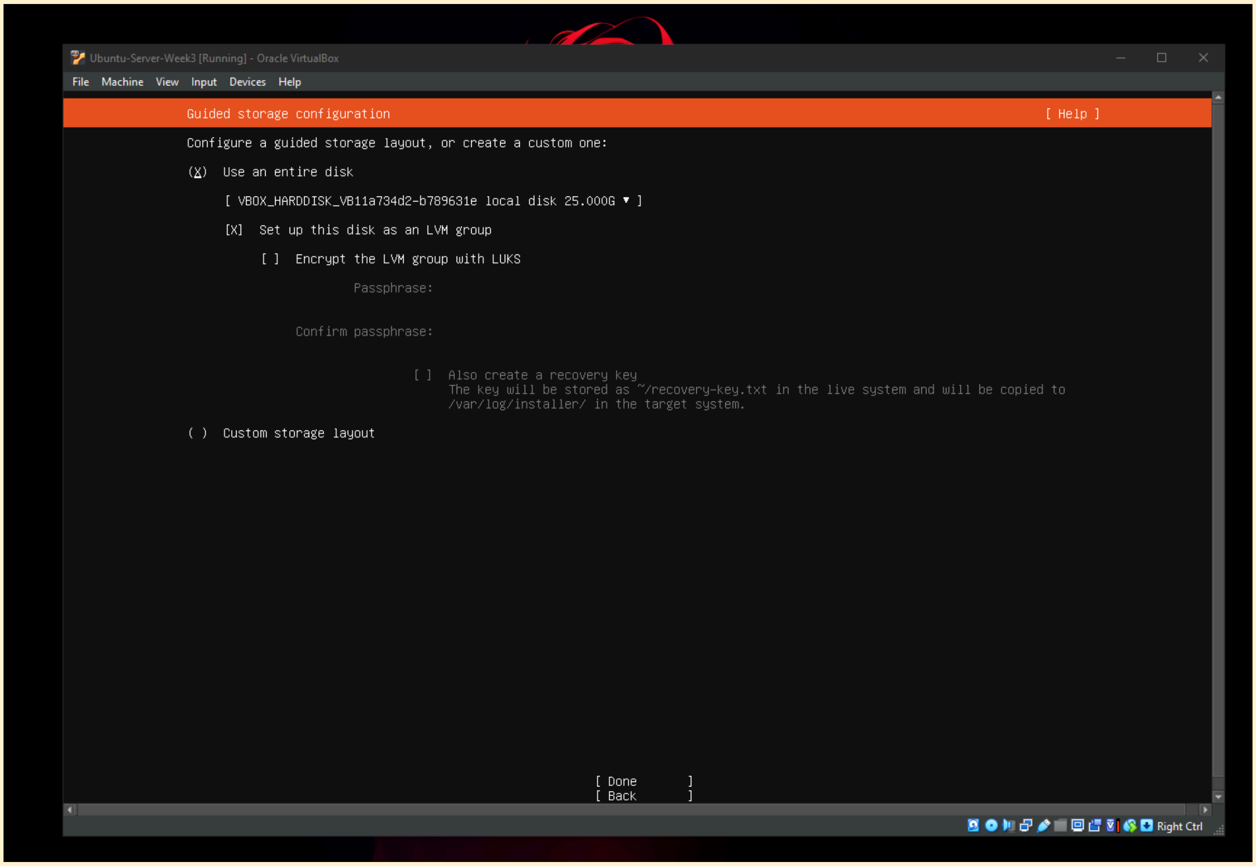
- Full Name: Arlou Anievas
- Username: anievas
- Password: [strong password — not documented for security]



5.6 Disk Partition

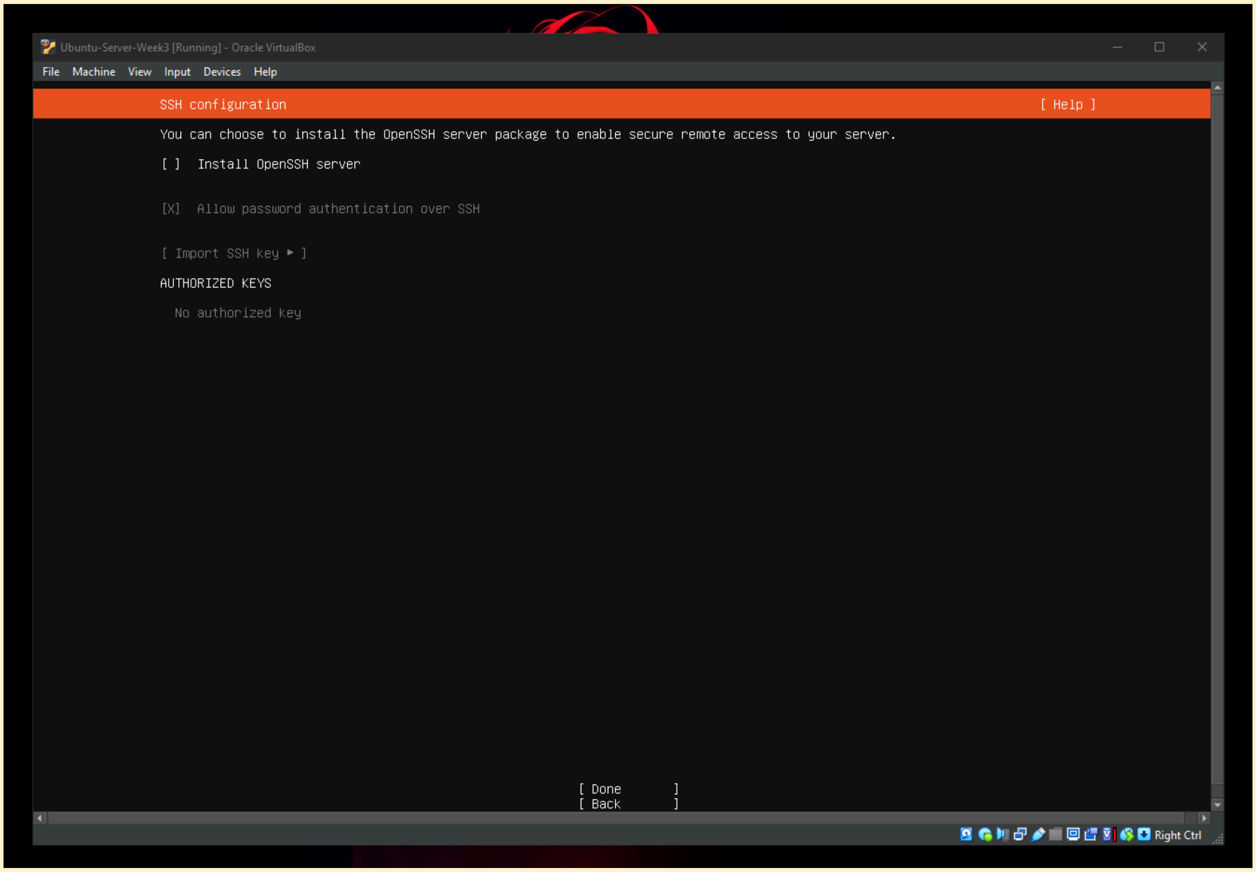
Partitioning method: Guided — Use Entire Disk

- Partition scheme: [e.g., single partition / LVM]
- File system used: [e.g., ext4]
- Disk size: 40 GB



5.7 SSH Server

OpenSSH Server was selected for installation to enable secure remote administration.

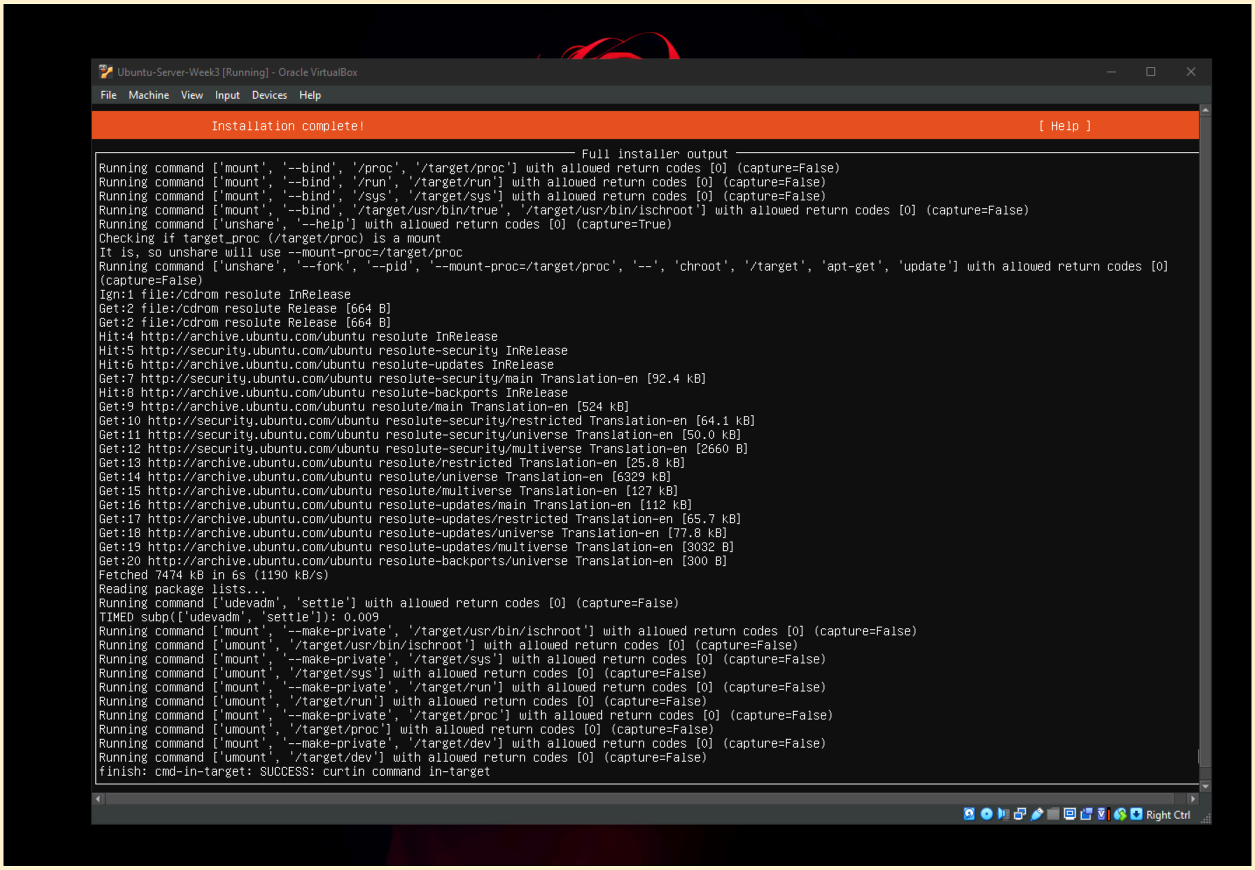


5.8 Additional Packages

No additional packages were installed, per project instructions.

5.9 Complete Installation

The installation completed successfully. The virtual machine was restarted and the installation ISO was removed from the optical drive when prompted.


Ubuntu-Server-Week3 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Installation complete!
[Help]
Full installer output
Running command ['mount', '--bind', '/proc', '/target/proc'] with allowed return codes [0] (capture=False)
Running command ['mount', '--bind', '/run', '/target/run'] with allowed return codes [0] (capture=False)
Running command ['mount', '--bind', '/sys', '/target/sys'] with allowed return codes [0] (capture=False)
Running command ['mount', '--bind', '/target/usr/bin/true', '/target/usr/bin/ischroot'] with allowed return codes [0] (capture=False)
Running command ['unshare', '--help'] with allowed return codes [0] (capture=True)
Checking if target_proc (/target/proc) is a mount
It is, so unshare will use --mount-proc=/target/proc
Running command ['unshare', '--fork', '--pid', '--mount-proc=/target/proc', '--', 'chroot', '/target', 'apt-get', 'update'] with allowed return codes [0] (capture=False)
Ign:1 file:/cdrom resolute InRelease
Get:2 file:/cdrom resolute Release [664 B]
Get:2 file:/cdrom resolute Release [664 B]
Hit:4 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:5 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:6 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [92.4 kB]
Hit:8 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:9 http://archive.ubuntu.com/ubuntu resolute/main Translation-en [524 kB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [64.1 kB]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [50.0 kB]
Get:12 http://security.ubuntu.com/ubuntu resolute-security/multiverse Translation-en [2660 B]
Get:13 http://archive.ubuntu.com/ubuntu resolute/restricted Translation-en [25.8 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute/universe Translation-en [6329 kB]
Get:15 http://archive.ubuntu.com/ubuntu resolute/multiverse Translation-en [127 kB]
Get:16 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [112 kB]
Get:17 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [65.7 kB]
Get:18 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [77.8 kB]
Get:19 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse Translation-en [3032 B]
Get:20 http://archive.ubuntu.com/ubuntu resolute-backports/universe Translation-en [300 B]
Fetched 7474 kB in 6s (1190 kB/s)
Reading package lists...
Running command ['udevadm', 'settle'] with allowed return codes [0] (capture=False)
TIMED subp(['udevadm', 'settle']): 0.009
Running command ['mount', '--make-private', '/target/usr/bin/ischroot'] with allowed return codes [0] (capture=False)
Running command ['umount', '/target/usr/bin/ischroot'] with allowed return codes [0] (capture=False)
Running command ['mount', '--make-private', '/target/sys'] with allowed return codes [0] (capture=False)
Running command ['umount', '/target/sys'] with allowed return codes [0] (capture=False)
Running command ['mount', '--make-private', '/target/run'] with allowed return codes [0] (capture=False)
Running command ['umount', '/target/run'] with allowed return codes [0] (capture=False)
Running command ['mount', '--make-private', '/target/proc'] with allowed return codes [0] (capture=False)
Running command ['umount', '/target/proc'] with allowed return codes [0] (capture=False)
Running command ['mount', '--make-private', '/target/dev'] with allowed return codes [0] (capture=False)
Running command ['umount', '/target/dev'] with allowed return codes [0] (capture=False)
finish: cmd-in-target: SUCCESS: curtin command in-target
Right Ctrl

6. Configuration Details

Summary of key configuration decisions made during deployment:

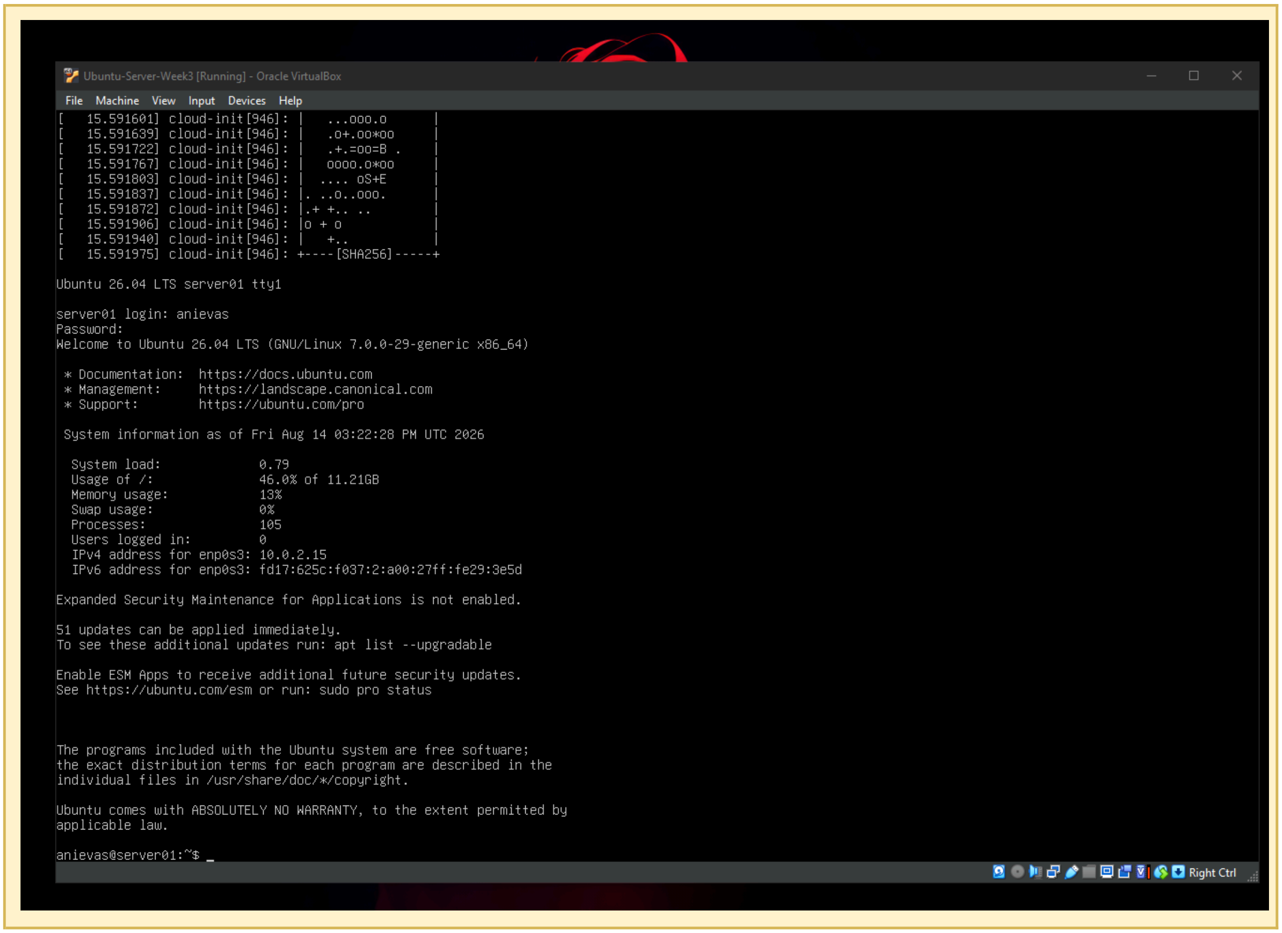
Setting	Value
Hostname	server01
Primary User	anievas (non-root, sudo-enabled)
Network Mode	NAT / DHCP
Partitioning	Guided — Entire Disk
SSH	Enabled (OpenSSH Server)
Additional Packages	None

7. Verification Commands

After installation, the server was verified using the following tasks and commands.

Task 1 — Login

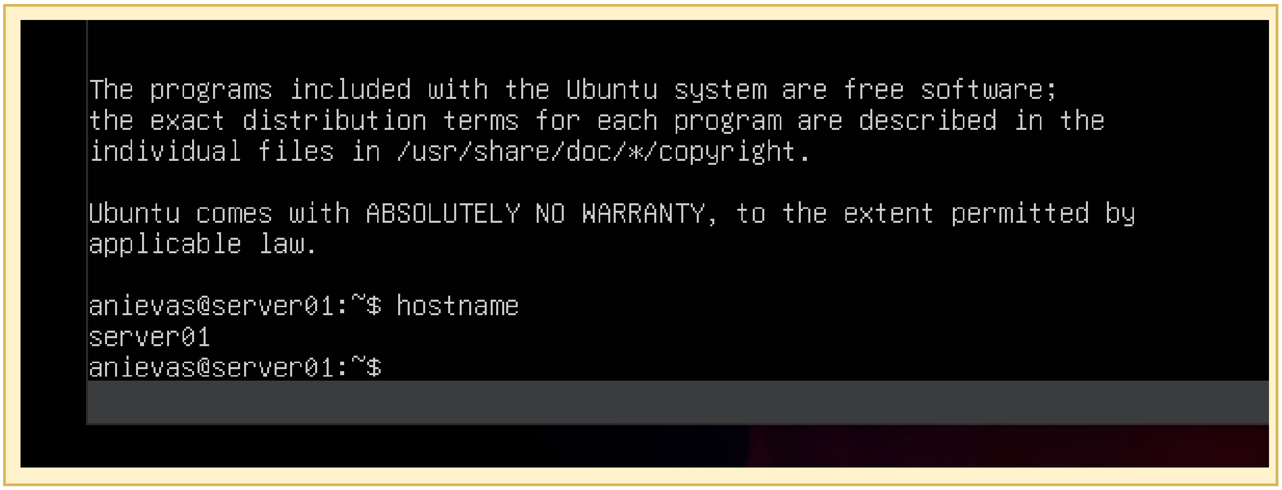
Logged in successfully using the created administrative account.



Task 2 — Verify Hostname

Command:

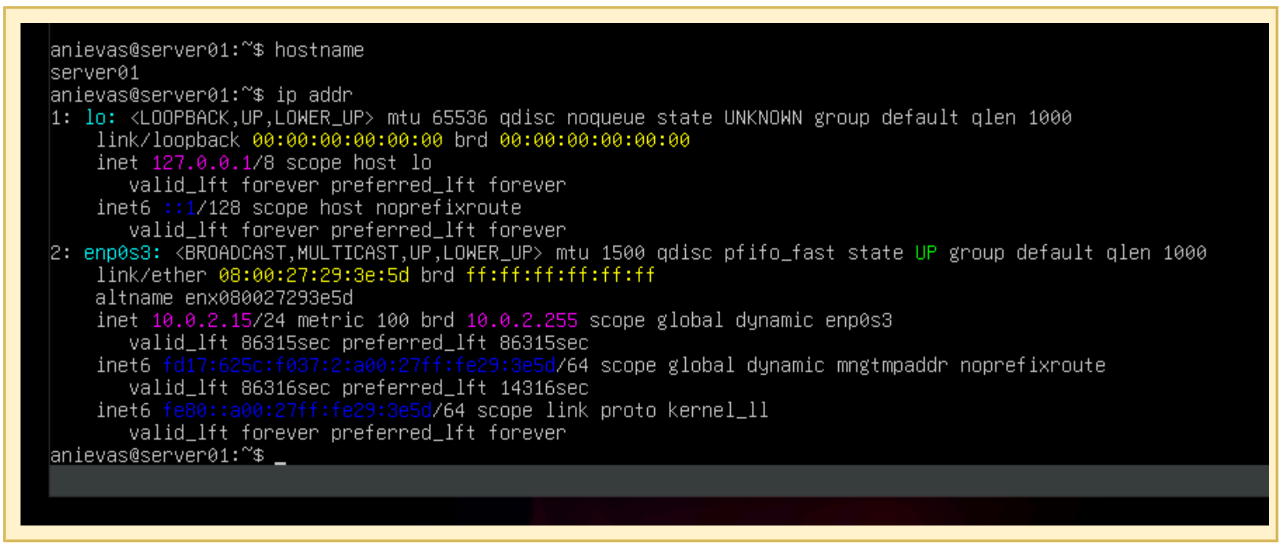
```
hostname
```



Task 3 — Verify IP Address

Command:

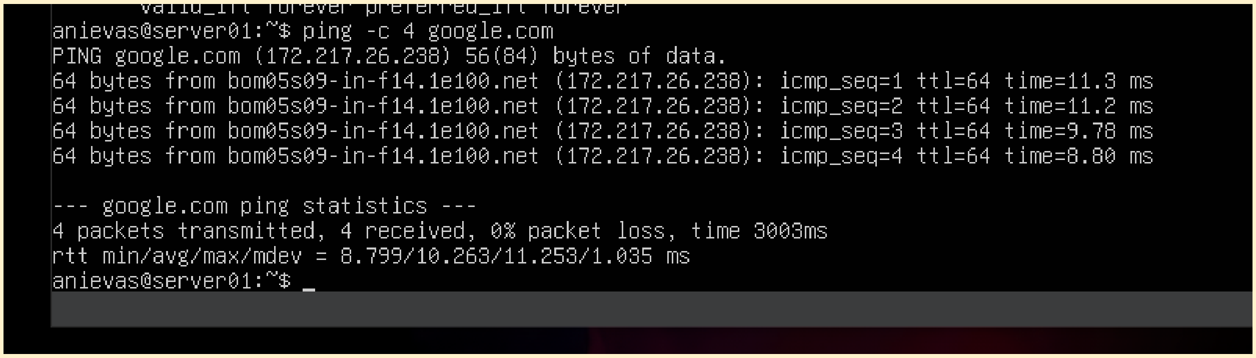
```
ip addr
```



Task 4 — Test Internet Connectivity

Command:

```
ping -c 4 google.com
```



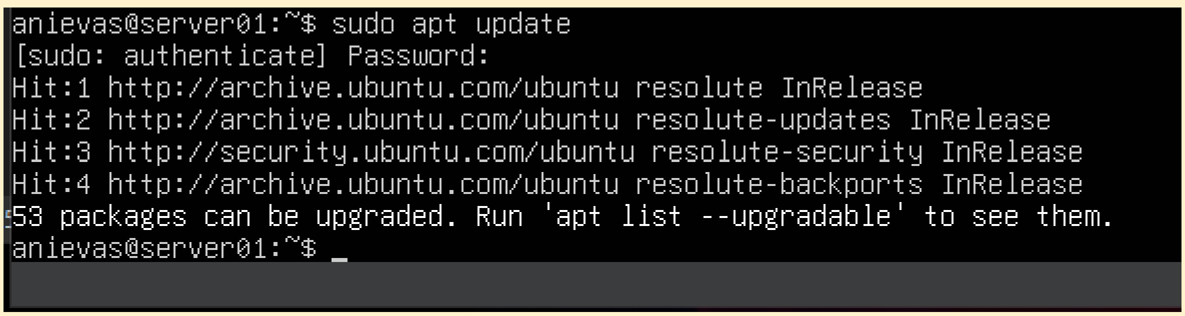
```
       valid_lft forever preferred_lft forever
anievas@server01:~$ ping -c 4 google.com
PING google.com (172.217.26.238) 56(84) bytes of data.
64 bytes from bom05s09-in-f14.1e100.net (172.217.26.238): icmp_seq=1 ttl=64 time=11.3 ms
64 bytes from bom05s09-in-f14.1e100.net (172.217.26.238): icmp_seq=2 ttl=64 time=11.2 ms
64 bytes from bom05s09-in-f14.1e100.net (172.217.26.238): icmp_seq=3 ttl=64 time=9.78 ms
64 bytes from bom05s09-in-f14.1e100.net (172.217.26.238): icmp_seq=4 ttl=64 time=8.80 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3003ms
rtt min/avg/max/mdev = 8.799/10.263/11.253/1.035 ms
anievas@server01:~$ _
```

Task 5 — Update the Server

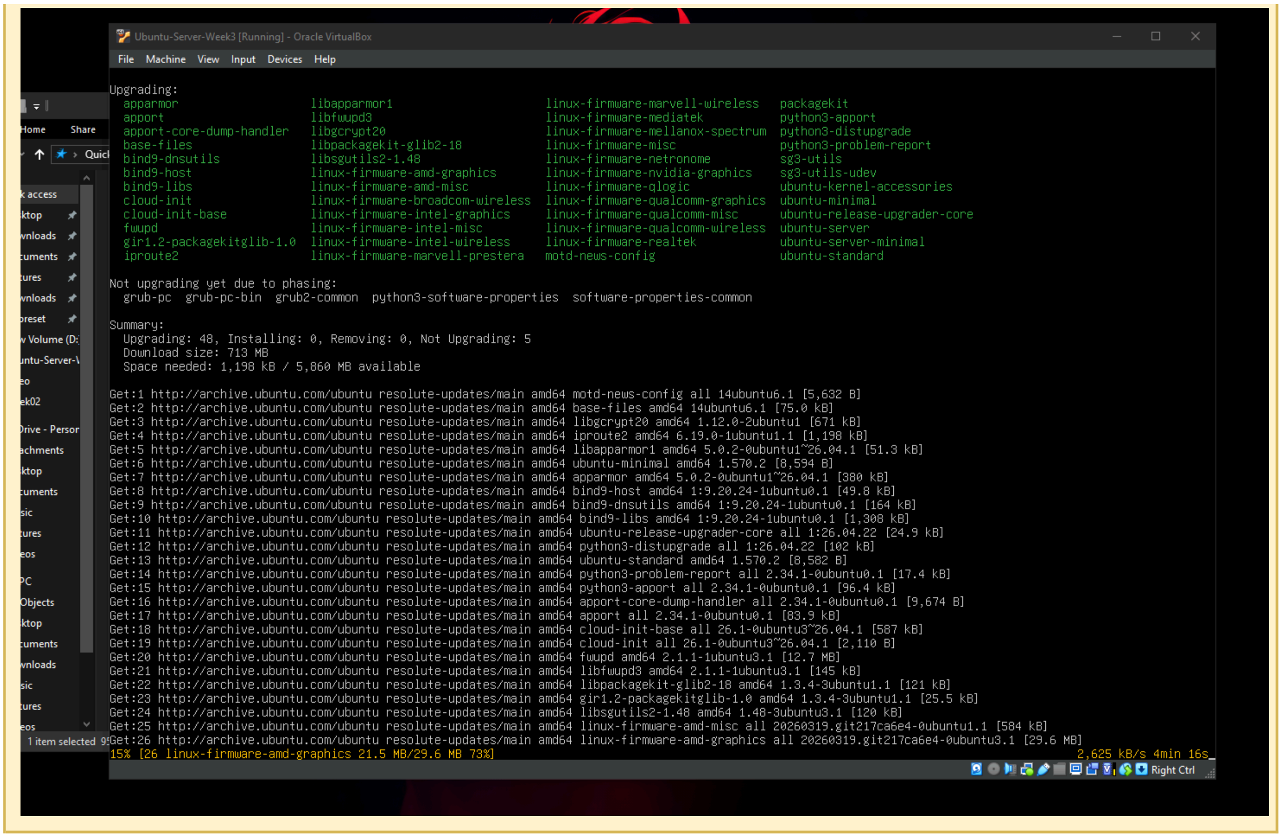
Commands:

`sudo apt update`

`sudo apt upgrade -y`



```
anievas@server01:~$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
53 packages can be upgraded. Run 'apt list --upgradable' to see them.
anievas@server01:~$ _
```

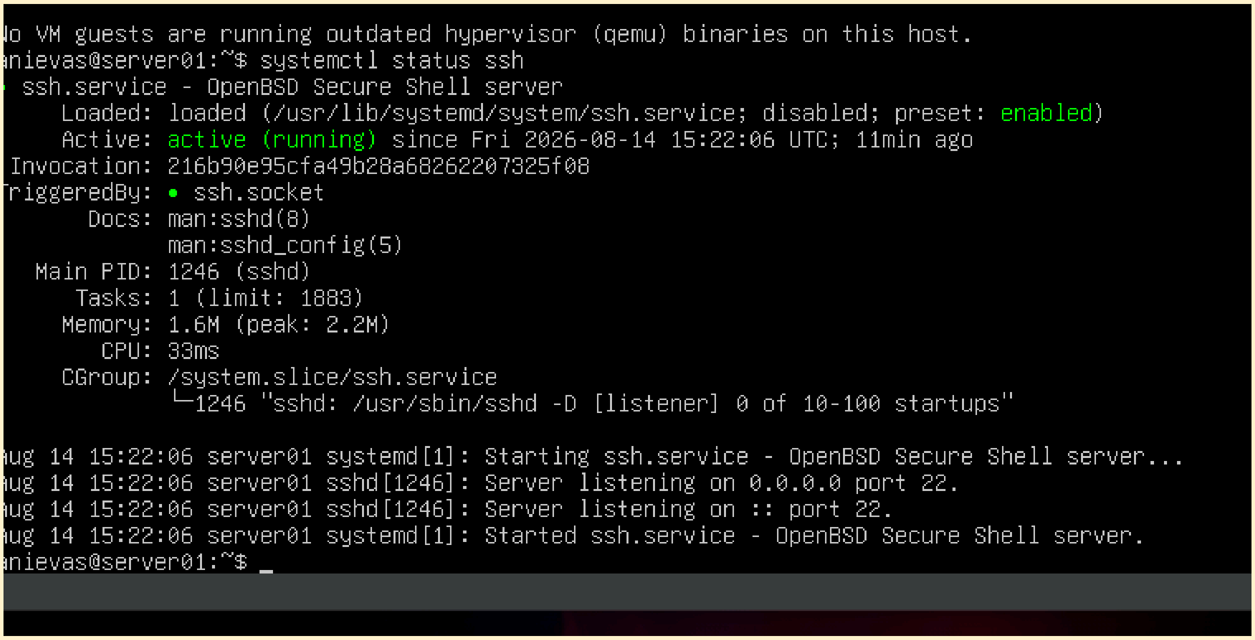


Task 6 — Verify SSH Service

Command:

```
systemctl status ssh
```

Expected result: active (running)



```
lo VM guests are running outdated hypervisor (qemu) binaries on this host.
anievas@server01:~$ systemctl status ssh
• ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
     Active: active (running) since Fri 2026-08-14 15:22:06 UTC; 11min ago
 Invocation: 216b90e95cfa49b28a68262207325f08
riggeredBy: • ssh.socket
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 1246 (sshd)
      Tasks: 1 (limit: 1883)
     Memory: 1.6M (peak: 2.2M)
        CPU: 33ms
     CGroup: /system.slice/ssh.service
             └─1246 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Aug 14 15:22:06 server01 systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Aug 14 15:22:06 server01 sshd[1246]: Server listening on 0.0.0.0 port 22.
Aug 14 15:22:06 server01 sshd[1246]: Server listening on :: port 22.
Aug 14 15:22:06 server01 systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
anievas@server01:~$ _
```

8. BIOS vs UEFI Comparison

Criteria	BIOS	UEFI
Definition	Basic Input/Output System — legacy firmware standard	Unified Extensible Firmware Interface — modern firmware standard
Boot Process	Runs in 16-bit real mode; reads Master Boot Record (MBR)	Runs in 32/64-bit mode; reads GPT partition table directly
Max Disk Support	Up to 2 TB (MBR limitation)	Supports drives larger than 2 TB (GPT)
Partition Style	MBR (max 4 primary partitions)	GPT (up to 128 partitions)
Security Features	Minimal; no native secure boot	Secure Boot, supports encryption & authentication
Speed	Slower boot due to sequential initialization	Faster boot via parallel device initialization
Advantages	Simple, widely compatible with legacy hardware	Faster, more secure, supports large drives, GUI support
Disadvantages	Slower, insecure, limited disk/partition support	More complex; legacy OS compatibility issues
Modern Usage	Largely phased out; legacy systems only	Standard on all modern PCs and servers

Why UEFI Has Largely Replaced BIOS

UEFI has become the industry standard for modern computers because it directly addresses the core limitations of legacy BIOS firmware. Where BIOS is restricted to the Master Boot Record and a maximum of four primary partitions on drives up to 2 TB, UEFI uses the GUID Partition Table (GPT), which supports drives far larger than 2 TB and up to 128 partitions — a necessity for enterprise servers handling large-scale storage.

Security is another major driver of adoption. UEFI's Secure Boot feature validates the digital signature of the bootloader and operating system before execution, helping prevent rootkits and other boot-level malware from compromising the system — a protection that BIOS simply cannot offer.

Performance also favors UEFI: its ability to initialize hardware components in parallel, rather than sequentially as BIOS does, results in noticeably faster boot times. UEFI additionally supports a graphical, mouse-driven interface and network-based diagnostics, improving usability for administrators managing firmware settings remotely.

For enterprise deployments such as the Ubuntu Server installation performed in this project, UEFI provides the security, scalability, and performance needed to reliably support modern workloads — which is why virtually all current server and desktop hardware ships with UEFI as the default firmware, with legacy BIOS mode retained only for backward compatibility.

9. Boot Process Flowchart

The diagram below illustrates the complete Ubuntu boot sequence, from power-on through to the login prompt.

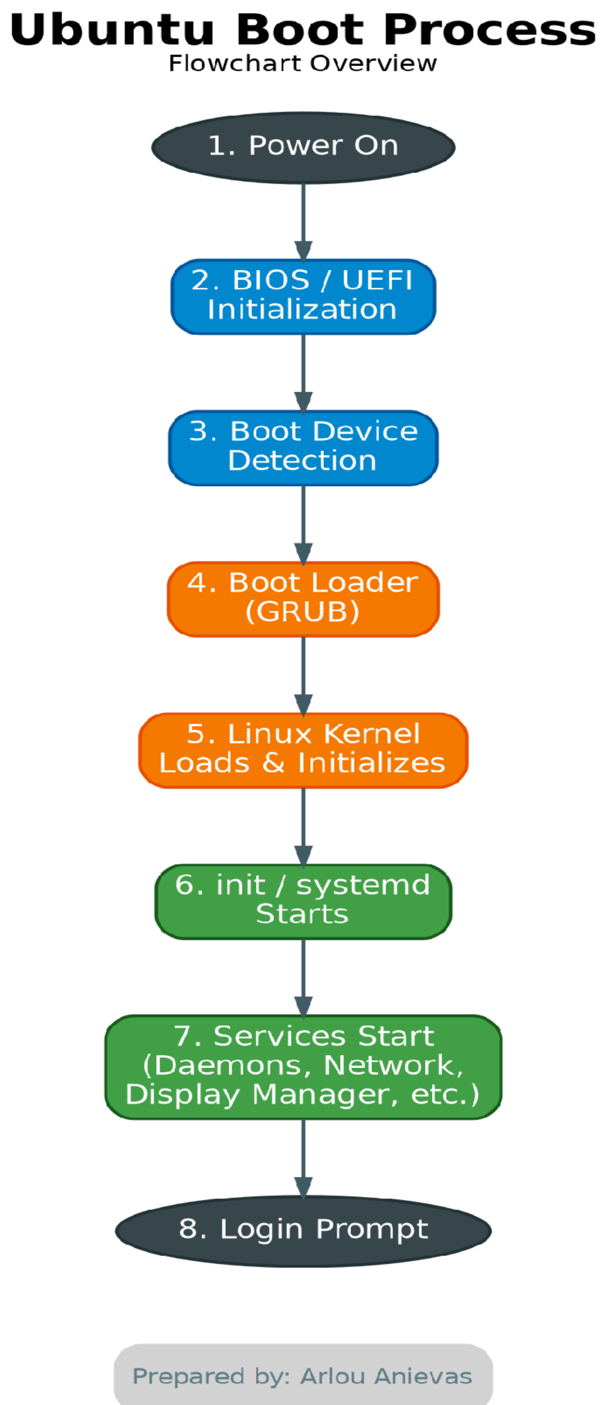


Figure 1. Ubuntu Boot Process Flowchart — Prepared by Arlou Anievas

Boot Stage Summary

1. Power On — Hardware receives electrical power; CPU begins executing firmware code.
2. BIOS/UEFI Initialization — Firmware performs POST (Power-On Self-Test) and initializes hardware components.
3. Boot Device Detection — Firmware locates a bootable device according to the configured boot order.
4. Boot Loader (GRUB) — GRUB loads, presents OS selection (if applicable), and hands off to the kernel.
5. Linux Kernel — The kernel loads into memory, initializes drivers, and mounts the root filesystem.
6. init/systemd — systemd (PID 1) starts as the first user-space process and begins managing the boot targets.
7. Services Start — systemd starts daemons, networking, SSH, and other configured services in parallel.
8. Login Prompt — The system presents a login prompt, indicating the server has booted successfully.

10. Windows Server Installation Summary

As part of the bring-home assignment, Windows Server Evaluation was installed in a separate virtual machine to support the operating system comparison in Section 11.

- Computer Name: [enter assigned computer name]
- Administrator Password: [configured — not documented for security]
- Installation Result: [e.g., completed successfully; logged in without issues]

11. Operating System Comparison

Criteria	Windows Server	Ubuntu Server	Rocky Linux
Licensing	Paid, per-core/CAL licensing	Free & open-source	Free & open-source
User Interface	GUI by default (Server Core optional)	CLI by default; GUI optional	CLI by default; GUI optional
Package Management	Windows Update / PowerShell	APT (.deb packages)	DNF/YUM (.rpm packages)
Security	Regular patches; larger attack surface historically	Strong community patching; AppArmor	Enterprise-grade; SELinux enabled by default
Performance	Higher resource overhead	Lightweight, efficient	Lightweight, RHEL-comparable performance
Enterprise Use Cases	Active Directory, .NET apps, Exchange	Web hosting, cloud, containers, DevOps	RHEL-compatible enterprise workloads
Advantages	Native AD/GUI integration, vendor support	Huge community, frequent updates, free	Long-term stability, RHEL binary compatibility
Disadvantages	Licensing cost, higher resource use	Shorter default support cycle options vary	Smaller community than Ubuntu

Discussion

Windows Server remains the preferred choice for organizations invested in Microsoft's ecosystem, particularly where Active Directory, Exchange, or .NET applications are core to operations — its licensing cost is offset by native integration and vendor support.

Ubuntu Server is well suited to web hosting, cloud infrastructure, and DevOps environments thanks to its free licensing, extensive package repository, and strong community support, as demonstrated in the deployment carried out for this project.

Rocky Linux appeals to enterprises requiring long-term stability and binary compatibility with Red Hat Enterprise Linux (RHEL), making it a common choice for regulated or mission-critical environments that previously relied on CentOS.

12. Problems Encountered

During the Ubuntu Server installation, I encountered a few problems while setting up the virtual machine. One of the main issues was that hardware virtualization was disabled on my computer. Because virtualization was turned off, the virtual machine could not run properly and I was unable to start the Ubuntu Server installation normally. I also had a problem with making sure that the Ubuntu Server ISO was properly connected to the virtual machine. At first, the virtual machine did not boot into the Ubuntu installer because the ISO was not selected correctly. Another issue was checking the network connection and identifying the assigned IP address. Since the network was configured using DHCP, I had to use the ip addr command to find the IP address assigned to the server. I also needed to make sure that the SSH service was installed and running after the installation.

13. Solutions Implemented

To solve the virtualization problem, I restarted my computer and entered the BIOS/UEFI settings. I located the virtualization setting and enabled it. After saving the changes and restarting the computer, I was able to run the virtual machine properly. For the ISO problem, I checked the virtual machine's storage settings and made sure that the Ubuntu Server ISO was properly attached to the optical drive. After restarting the virtual machine, the Ubuntu installation menu appeared normally. For the network connection, I checked the virtual machine's network adapter and made sure it was enabled and configured to use NAT. I then used the ip addr command to verify that the server received an IP address through DHCP. For SSH, I made sure that the OpenSSH Server option was selected during installation. After the installation, I used systemctl status ssh to confirm that the SSH service was active and running.

14. Lessons Learned

This activity helped me understand that installing an operating system on a server requires more than simply following the installation wizard. I learned that the virtual machine must be properly prepared before installation, including checking the CPU, RAM, storage, network settings, ISO file, and hardware virtualization. One of the most important things I learned was how BIOS/UEFI settings can affect virtualization. Since virtualization was disabled on my computer, I had to enter the BIOS/UEFI settings and enable it before I could properly run the Ubuntu Server virtual machine. This helped me understand the role of firmware in preparing hardware before an operating system starts. I also learned some basic Linux system administration commands such as hostname, ip addr, ping, apt update, and systemctl status ssh. These commands helped me verify that the server was properly configured and working. Overall, the project improved my confidence in deploying Ubuntu Server and troubleshooting common installation problems. I also learned that proper documentation is important because it allows other administrators to understand the configuration and solutions used when problems occur.

15. References

Ubuntu Server Documentation. Canonical Ltd. https://ubuntu.com/server/docs

Microsoft Windows Server Documentation. Microsoft Corp. https://learn.microsoft.com/windows-server/

Rocky Linux Documentation. Rocky Enterprise Software Foundation. https://docs.rockylinux.org/

UEFI Specification. Unified Extensible Firmware Interface Forum. https://uefi.org/specifications

GRUB Manual. GNU Project. https://www.gnu.org/software/grub/manual/grub/

systemd Documentation. freedesktop.org. https://www.freedesktop.org/wiki/Software/systemd/